

Extendable multirate real-time simulation of active distribution networks based on field programmable gate arrays



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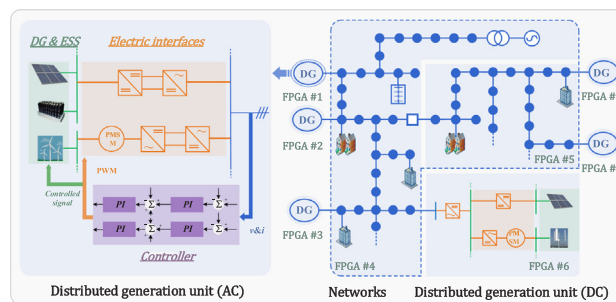
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HIGHLIGHTS

- The real-time simulation framework for active distribution networks was built.
- A multirate simulation method was proposed with high-accuracy and stability.
- The detailed hardware design of the real-time simulator was presented.

GRAPHICAL ABSTRACT



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ABSTRACT

Real-time simulation of large-scale active distribution networks exhibiting a wide range of time-scales puts forward higher requirements for simulation accuracy and efficiency. This paper presents an extendable method and design for the real-time simulation of active distribution networks utilising high-performance hardware field programmable gate arrays. In the aspect of numerical algorithm, a high-accuracy and stable multirate simulation algorithm is proposed. The entire active distribution network is decoupled into different subsystems by their inherent time-scales and distinct time steps are used to solve the subsystems. Then root-matching method is adopted to form the exponential difference equations that represent the behaviours of the electric distribution network being modelled, which eliminates the truncation errors and thus provides a highly accurate time-domain solution. To handle the interface between the subsystems, a multirate interfacing method is proposed. The hardware design of the multirate interfaces is presented as well. With the multirate algorithm, a fully functional extendable real-time simulator based on field programmable gate arrays is designed and implemented. Two modified IEEE 33-node systems with photovoltaics and energy storage are simulated on the 4-field-programmable-gate-arrays-based real-time simulator. Simulation results are compared with the commercial simulation tool PSCAD/EMTDC to validate the correctness and effectiveness of the proposed method and design.

1. Introduction

With the increasing penetration of distributed generators (DGs), mainly based on renewable energy sources (RESs), electric power

distribution systems have been converted from passive to active distribution networks (ADNs) [1,2]. The transition has brought profound and continuous changes in system architecture and operation modes [3,4]. To effectively control and manage ADNs, plenty of novel system

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Nomenclature

Parameters

G	Nodal conductance matrix of the overall system
G_s	Node conductance matrix of the slow subsystem
G_f	Node conductance matrix of the fast subsystem

Variables

i	Injected node currents of the overall system
v	Node voltages of the overall system

I_{hist}	History terms of the overall system
i_s, i_f	Injected node currents of the slow/fast subsystem
v_s, v_f	Nodal voltages of the slow/fast subsystem
$I_{h,s}, I_{h,f}$	History current sources of the slow/fast subsystem except for the Bergeron line models
$I_{IF,s}, I_{IF,f}$	History current sources of the Bergeron line models belonging to the slow/fast subsystem
$i_{IF,s}, i_{IF,f}$	Branch currents of the Bergeron line models belonging to the slow/fast subsystem
$v_{IF,s}, v_{IF,f}$	Nodal voltages of the Bergeron line models belonging to the slow/fast subsystem

equipment and operation strategies, ranging from DG controllers to system-level voltage & frequency regulation strategies, have been developed and integrated into the ADN. Before coming to use, these novel equipment and strategies have to be validated in the hardware-in-the-loop testbed. The studied ADN needs to be completely or partly emulated by real-time simulators to reproduce the detailed dynamic behaviours at discrete time intervals strictly within the same time in real-world clock [5]. But the dynamics in a typical ADN can be as fast as tens of microsecond level [6]. These requirements highlight the importance of developing the simulators that can perform large-scale and high-accuracy real-time simulation of ADNs.

Due to the expanding scale and growing time span of system transients brought by renewable energy integration, real-time simulation of ADNs is challenging [7]. Considering the limited computational resources, system scale and calculation speed are the two conflicting requirements for real-time simulators. The system modelling for ADNs covers not only the circuits of DGs, energy storage systems (ESSs) and power electronic devices but also their controllers and system-level voltage & frequency regulators. The detailed modelling of these components enlarges the simulation scale and thus entails excessive computational resources. In order to reproduce the transients with high fidelity, the simulation time-step in microsecond level, generally determined by the fastest dynamics, is required. To meet the critical real-time constraints, massive computational resources have to be paid to exchange the rapidity of transient simulation. The requirements for both large-scale and high-accuracy real-time simulation of ADNs aggravates the computation burden.

Some commercial real-time simulators have been developed and applied to power and energy system studies [6]. These simulators tend to perform the real-time simulation of relatively smaller system scale with specific research field [8–11]. For the system-level modelling and simulation with DGs and ESSs, these simulators often suggest employing equivalent simplification techniques and larger time-steps to reduce the problem scale and lower computational burden. As a result, the high-frequency transients such as switching surges and harmonics of power electronic devices probably won't be captured [12,13].

The advanced real-time simulator, e.g. RTDS®, RT-LAB®, HYPER-SIM®, is making an effort to realise the simulation of larger system scales in longer time span while preserving the fidelity of high-frequency transients. If the entire system is solved using the same small time-step associated with the fast-varying components, computational resources will be wasted since the time-step may be unnecessary for other parts of the system. While if large time-step associated with slow-varying components is adopted, the dynamics of fast-varying components may not be captured and numerical problems may also arise. By decoupling into different time-scale subsystems, multirate simulation provides a feasible way to resolve the conflicts. Accordingly, distinct time-steps can be used for each subsystem and the consumption of computational resources will be reduced.

Another consideration is the structure and parallelism of the underlying hardware to support the simulator. In some traditional real-

time simulators, the core computational resources are based on sequential processors, such as reduced instruction set computers (RISCs), digital signal processors (DSPs), central processing units (CPUs), and PC clusters [6]. 'Sequential' means that the instructions in these processors are executed one after another, which heavily limits the improvement of calculation speed. Field programmable gate array (FPGA) offers an alternative to speed up the instruction execution. It has an intrinsic massive parallel architecture that can be partitioned and configured into a large number of parallel processing units. Another advantage of FPGA in hardware performance is the pipelining processing. With these distinctive features, FPGA is capable of realising real-time circuit simulation in nanosecond level [14]. Note that FPGA has been introduced as co-processors to emulate power electronic devices in RTDS and RT-LAB [6].

A number of research efforts have been conducted to accelerate the transient simulation from the aspect of algorithm development and hardware improvement. Ref. [15] promoted a multirate simulation method based on the waveform relaxation (WR) technique. Iterations are performed until the solutions at all interfacing points are matched. As the numbers of iterations are uncertain with different solution points, it imposes limitations particularly in the context of real-time simulation. In Ref. [16], a multirate method based on mixed Adams-Backward Differentiation Formula (BDF) scheme was proposed. The time-steps for both the fast and slow variables are changing during the simulation, which is generally considered to be unsuitable for real-time simulation. The multirate method in [17] was based on the concept of interlinked Multi-Area Thevenin Equivalents (MATE) without iterations. Considering that the solutions for subsystems are performed in serial, the calculation speed is relatively decreased. The subsequent Ref. [18] proposed a MATE-based multirate simulation method for electric networks with mixed integration rules. Ref. [19] implemented the real-time simulation of a small-scale traditional power system based on a single FPGA. In Ref. [20], a hardware design for the real-time simulation of traditional power systems based on multiple FPGAs was proposed. In the design, FPGAs are embedded on a commercial integration board which limits the extendibility of the simulator. FPGAs have also been employed to perform the real-time control algorithms for the hardware-in-the-loop test especially in the field of power and energy. In Ref. [21], FPGA was used to implement the proposed control technique that aimed to minimize the hydrogen consumption in polymer electrolyte membrane fuel cell systems. Ref. [22] realized a built-in-house electronic control system on an embedded FPGA for studying the spray characteristics of the wide distillation fuel. An in-house developed FPGA-based open engine controller in Ref. [23] was used to change the experimental conditions in testing the injection strategies for improving the performance of the butanol-diesel common rail dual fuel engine.

Although some simulators have realised the parallel multirate simulation for bulk power systems using the natural travel time delay of transmission lines [24], realising the real-time simulation of ADNs is still challenging. Some simulators have been essentially designed and developed mainly facing the electric networks with fewer power

electronic devices and the computational resource consumption is affordable. The requirement for hardware performance is less strict than only the power electronic devices are modelled and simulated on FPGAs. For ADNs including massive DGs, the computation burden, mainly resulting from the detailed modelling of DGs, is relatively heavier than extendable and powerful real-time simulators are required.

To design a high-performance simulator facing the increasing complexity and real-time demands, difficulties lie in the high-accuracy numerical algorithms and appropriate hardware design and realisation. When performing the multirate simulation, numerical oscillations may arise from the original transient simulation method exhibiting truncation errors [25]. To eliminate the oscillations, more effective and accurate numerical algorithms should be developed. Besides, the interfaces between subsystems should be handled properly since the simulation accuracy and stability depend on the multirate interfaces significantly. For the hardware design, efforts are needed to analyse the simulation procedure, find the potential parallelism, design functional hardware modules and connect them.

Considering the above issues, this paper presents an extendable multirate real-time simulation method for ADNs based on FPGAs. Firstly, the entire ADN is physically decomposed into network components and distributed generation units according to their time-scales, and interfaced by the propagation delay time of transmission lines. When solving the network components, an exponential form of difference equations that characterize the network behaviours is derived based on root-matching method [26]. It does not suffer from truncation errors, and yet provides a highly efficient and accurate time-domain solution especially for large simulation time-step. Then, the multirate interfacing algorithm, i.e. interpolation and averaging operations for the interface variables, is performed to deal with the interface between the subsystems with distinct time-scales. Finally, the above simulation algorithms are translated into hardware language and performed on FPGA. The subsystems are allocated to separate FPGAs and solved in parallel with individual time-step sizes. To validate the proposed approach and design, real-time simulation of two modified IEEE 33-node systems with photovoltaics (PVs) and energy storage is implemented on a 4-FPGA-based simulator. Moreover, different time-steps for the slow-varying network are adopted to validate the stability of the multirate interface. Simulation results are compared with PSCAD/EMTDC to verify the proposed simulator.

The major contributions are summarized as follows: (i) An extendable real-time simulation framework for ADNs is established. The modelling method for both the network components and distributed generation units are described. (ii) A multirate interfacing method is presented to deal with the interface between the subsystems. (iii) The

detailed hardware design of the multi-FPGA based extendable real-time simulator is implemented. In the simulator, several FPGAs operate with specified time-steps synchronously.

The rest of this paper is organised as follows. Section 2 introduces the real-time simulation framework and system modelling of ADN. Section 3 describes the proposed multirate simulation method and the detailed hardware design of the FPGA-based real-time simulator. In Section 4, real-time simulation of the modified IEEE 33-node systems with PVs and energy storage is implemented. Conclusions are stated in Section 5.

2. Extendable real-time simulation framework

A complete ADN is mainly composed of network components and distributed generation units as Fig. 1 shows. Network components refer to lines, transformers, shunt capacitors and other system-level regulation devices. Distributed generation units consist of various energy conversion and storage devices, electric interfaces (e.g. power electronic converters and electric motors) and their controllers. The time constants associated with the high-frequency power electronic devices in distributed generation units are on the order of tens of microseconds [27], while the dynamics in network components are much slower. For this reason, ADNs can be partitioned into the electric networks and distributed generation units. By decoupling ADNs with the subsystem partitioning, multirate simulation is possible to be conducted.

Although the multirate simulation constitutes an efficient solution, for the network discretizing with a large integration step, its numerical solutions are less accurate than that using relative small integration step. The reason is that the trapezoidal integration rule, generally applied to form the difference equations of network components, is based on a truncated Taylor series and the resulting truncation errors increase with the integration step. To eliminate the error and provide a highly accurate time-domain solution, an exponential form of the difference equation has been developed based on root-matching method [26].

2.1. Modelling of network components with the root-matching method

Network components of ADNs, e.g. loads, filters, shunt capacitor and ground resistance are generally modelled by linear elements such as resistance, inductance, capacitance and their series or parallel combinations. The dynamic behaviours of these elements can be represented mathematically by a series of algebraic differential equations (DAEs). By discretizing the DAEs and solving the resulting difference equations, the dynamic behaviours can be reproduced and thus the time-domain solutions are achieved. The steps followed to form and

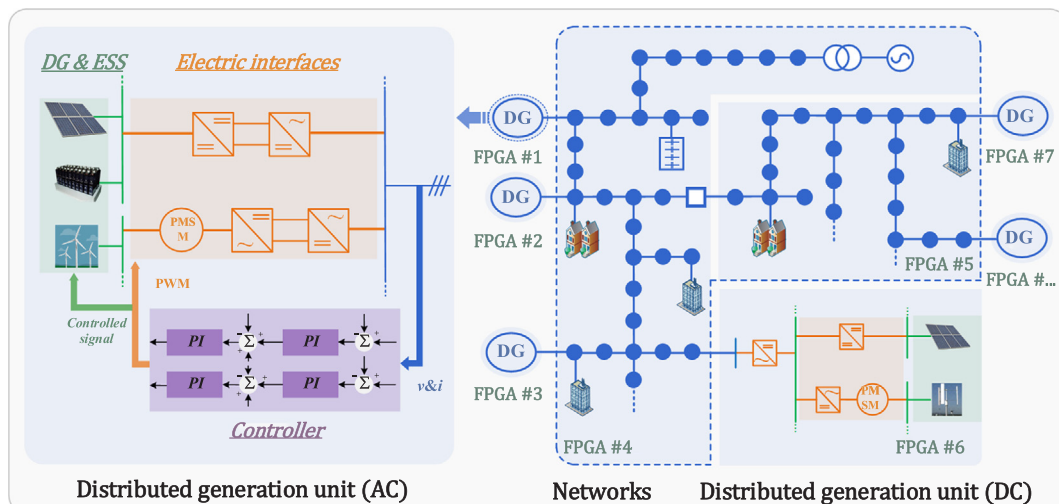


Fig. 1. System partition of active distribution networks.

solve the difference equations based on the root-matching method are as follows.

- (1) Discretize the branch DAE using the root-matching method and form its exponential difference equation. The series resistance-inductance branch in Fig. 2(a), is taken as an example. The differential equation is represented by

$$v_{km}(t) = v_k(t) - v_m(t) = L \frac{di_{km}(t)}{dt} + Ri_{km}(t) \quad (1)$$

in which v_k and v_m are the node voltages, v_{km} and i_{km} are the branch voltage and branch current. Performing Laplace transformation for $v_{km}(t)$ and $i_{km}(t)$, yields the transfer function $H(s)$ as

$$H(s) = \frac{I_{km}(s)}{V_{km}(s)} = \frac{1}{Ls + R} \quad (2)$$

Representing $1/R$ and L/R by G and τ respectively, $H(s)$ can be expressed as

$$H(s) = \frac{G}{1 + s\tau} \quad (3)$$

To discretize $H(s)$, the transfer function $H(z)$ in the z -plane is written by using the mapping $z = e^{s\Delta t}$ (Δt is the time-step), thus ensuring its poles and zeros in the same place with $H(s)$. Also, a constant k is added to allow adjustment of the final value. The transfer function $H(z)$ is depicted by (4).

$$H(z) = \frac{kz}{z - e^{-\Delta t/\tau}} \quad (4)$$

Applying Final Value Theorem to compute the final value of $H(s)$ for a unit step input yields

$$\lim_{s \rightarrow 0} \{s \cdot H(s) \cdot E(s)\} = G \quad (5)$$

The final value of $H(z)$ for unit step input is computed as

$$\lim_{z \rightarrow 1} \{(z-1) \cdot H(z) \cdot E(z)\} = \frac{k}{1 - e^{-\Delta t/\tau}} \quad (6)$$

The constant k is adjusted to be the correct value.

$$k = G(1 - e^{-\Delta t/\tau}) \quad (7)$$

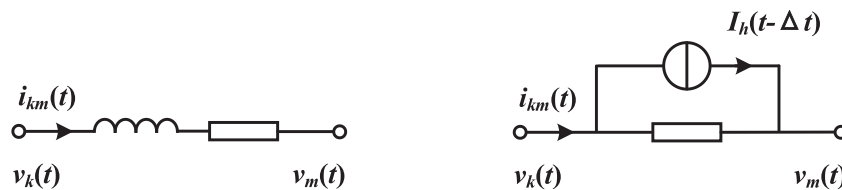
By rearranging the resulting z -domain equation, the difference equation in exponential form for RL branch is obtained as

$$i_{km}(t) = \frac{(1 - e^{-\Delta t R/L})}{R} v_{km}(t) + e^{-\Delta t R/L} i_{km}(t - \Delta t) \quad (8)$$

Eq. (8) can be expressed as a Norton equivalent circuit illustrated in Fig. 2(b). It is composed of a conductance $(1 - e^{-\Delta t R/L})/R$ in parallel with a history current source.

To eliminate the numerical oscillations, the difference equations in the exponential form should be implemented for all resistances, inductances, capacitances and their combinations in series and parallel. In this paper, the difference equations of these components are organized into a unified format:

$$i(t) = G_{eq} v(t) + A_1 v(t - \Delta t) + A_2 i(t - \Delta t) + A_3 i(t - 2\Delta t) \quad (9)$$



(a) Series resistance-inductance branch.

(b) Equivalent circuit model.

Fig. 2. Series resistance-inductance branch and its equivalent circuit for real-time simulation.

The corresponding coefficients A_1, A_2, A_3 and equivalent conductance G_{eq} are given in Appendix A.

- (2) With all network components represented by Norton equivalents, the nodal equation can be established as

$$Gv(t) = i(t) - I_{hist} \quad (10)$$

In (10), I_{hist} is the known column vector made up of equivalent current sources. After solving this equation, the nodal voltages will be obtained.

2.2. Modelling of the distributed generation unit

The aforementioned principles and methods are aimed to solve the linear network components, constituting the electrical system solving module, which will be introduced in Section 3.3. For distributed generation units, there existing various nonlinear components, such as DGs, energy storage, power electronic devices and their controllers, the modelling approach is different. These nonlinear components fall into two categories, including components with continuous and discontinuous characteristics.

For the discontinuous nonlinear component (e.g. power electronic circuit), it is often represented by a zero resistance in closing state and an infinitely large resistance in opening state in off-line simulator [28]. However, a number of possible circuit topologies will result in re-factorizing the nodal conductance matrix, which may consume plenty of simulation time and be unsuitable for real-time simulation. This paper adopts the associated discrete circuit (ADC) to represent power electronic circuits [29]. The circuit in the closing state is represented by an inductance, while in the opening state it is represented by a series resistance-capacitance circuit. By adjusting the values of the parameters, the two equivalent conductances of the inductance and the resistance-capacitance series circuit will be equal that the system conductance matrix will remain unchanged.

For the continuous nonlinear components (e.g. DGs and controllers), the compensation-based method is usually used to avoid the solving of massive nonlinear equations [28]. The nonlinear component is represented by a current injection which is super-imposed on the linear system after a solution without the nonlinear element has first been found. Although the compensation-based method reduces the numbers of the nonlinear equations, solving the remaining equations still brings difficulties in the hardware design realisation due to the computational burden. In this paper, these components are modelled in the control system solving module. Unlike the electrical module, the control module provides a variety of functional blocks, such as the transfer function, trigonometric function, exponential function and so on. The blocks are represented by input-output relations, which will be introduced in Section 3.3.

3. Multirate simulation algorithm and hardware design

Multirate simulation of ADNs aims to decompose the entire system into subsystems with distinct time-scales and attain the correct time-domain responses with the corresponding time-steps. To show how the decomposition is performed and the interface between the subsystems

is dealt with, two subsystems are considered throughout this section: one representing the fast subsystem and the other representing the slow one as Fig. 3 shows. The two subsystems are connected by r short lines. The item ' i ' and ' j ' depict the nodes in the slow and fast subsystem respectively where the lines are connected. The integration step for the slow subsystem is ΔT and the fast one is Δt .

To decouple the two subsystems, this paper employs the Bergeron line model [30] to represent the short lines. As the solutions of the two subsystems are updated with different rates, the contribution by one subsystem for the other should be evaluated. In this section, the network decomposition and multirate interfacing method are presented in detail. On this basis, the hardware design of the FPGA-based extendable real-time simulator is implemented.

3.1. Network decomposition

Generally, the Bergeron line model can't be directly used for the modelling of the short line as the propagation delay time is often not longer than the time-step in ADN. In this paper, the short line is adjusted to have a propagation delay of one or several certain time-steps [31]. The short line is essentially modelled by a series resistance-inductance branch as Fig. 4(a) shows, where R and L are the lumped resistance and inductance. Assuming that the branch comprises infinitely connected inductances and resistances ($\Delta x \cong 0$), and adding unreal distributed capacitances throughout the line, a Bergeron line model is derived as Fig. 4(b) shows, where L' is the inductance, R' is the resistance and C' the capacitance per unit length. The length of the line is set to d artificially.

The characteristic equations of the Bergeron line model at both ends k and m are

$$\begin{cases} i_{km}(t) = 1/(Z + R/4)v_k(t) + I_k(t-\tau) \\ i_{mk}(t) = 1/(Z + R/4)v_m(t) + I_m(t-\tau) \end{cases} \quad (11)$$

The surge impedance Z and propagation delay time τ are given by

$$Z = \sqrt{L'/C'} \quad (12)$$

$$\tau = d\sqrt{L'C'} \quad (13)$$

The history current source I_k and I_m are computed as

$$\begin{cases} I_k = (B_1 v_k + B_3 i_{km}) + (B_2 v_m + B_4 i_{mk}) \\ I_m = (B_1 v_m + B_3 i_{mk}) + (B_2 v_k + B_4 i_{km}) \end{cases} \quad (14)$$

where the coefficients B_1 , B_2 , B_3 and B_4 are given by

$$\begin{aligned} B_1 &= \frac{-R/4}{(Z + R/4)^2}, B_2 = \frac{-Z}{(Z + R/4)^2}, B_3 = \frac{-R/4(Z - R/4)}{(Z + R/4)^2}, B_4 \\ &= \frac{-Z(Z - R/4)}{(Z + R/4)^2} \end{aligned} \quad (15)$$

Fig. 4(c) shows the resulting two-port equivalent circuit model. There is no direct connection between the two terminals, thus decomposition can be performed. Note that C' is a direct consequence of modelling the

RL branch as a Bergeron line model and would not normally exist in the circuit. It shouldn't be too large. In general, the value of X_c should be significantly larger than the surrounding impedances.

Applying the decomposition approach, the schematic network shown in Fig. 3 can be substituted by Fig. 5.

The nodal formations for the decoupled slow and fast subsystems using Bergeron line model are derived from (10) as

$$\begin{cases} G_s v_s(t) = i_s(t) - I_{h,s}(t - \Delta T) - I_{f,s}(t - \tau) \\ G_f v_f(t) = i_f(t) - I_{h,f}(t - \Delta t) - I_{f,f}(t - \tau) \end{cases} \quad (16)$$

where $I_{f,s}(t - \tau)$ and $I_{f,f}(t - \tau)$ are computed by

$$\begin{cases} I_{f,s}(t - \tau) = (B_1 v_{f,s}(t - \tau) + B_3 i_{f,s}(t - \tau)) + (B_2 v_{f,f}(t - \tau) + B_4 i_{f,f}(t - \tau)) \\ I_{f,f}(t - \tau) = (B_1 v_{f,f}(t - \tau) + B_3 i_{f,f}(t - \tau)) + (B_2 v_{f,s}(t - \tau) + B_4 i_{f,s}(t - \tau)) \end{cases} \quad (17)$$

To simplify the representation of (17), we have

$$\begin{cases} I_{f,s}(t - \tau) = I_{s,s}(t - \tau) + I_{s,f}(t - \tau) \\ I_{f,f}(t - \tau) = I_{f,f}(t - \tau) + I_{f,s}(t - \tau) \end{cases} \quad (18)$$

in which

$$\begin{cases} I_{s,s}(t - \tau) = B_1 v_{f,s}(t - \tau) + B_3 i_{f,s}(t - \tau) \\ I_{s,f}(t - \tau) = B_2 v_{f,f}(t - \tau) + B_4 i_{f,f}(t - \tau) \\ I_{f,f}(t - \tau) = B_1 v_{f,f}(t - \tau) + B_3 i_{f,f}(t - \tau) \\ I_{f,s}(t - \tau) = B_2 v_{f,s}(t - \tau) + B_4 i_{f,s}(t - \tau) \end{cases} \quad (19)$$

Hence, (16) can be rearranged as

$$\begin{cases} G_s v_s(t) = i_s(t) - I_{h,s}(t - \Delta T) - I_{s,s}(t - \tau) - I_{s,f}(t - \tau) \\ G_f v_f(t) = i_f(t) - I_{h,f}(t - \Delta t) - I_{f,f}(t - \tau) - I_{f,s}(t - \tau) \end{cases} \quad (20)$$

As can be seen from (20), when solving the slow subsystem, $I_{s,s}$ is updated with ΔT and $I_{s,f}$ is updated with Δt . If τ is not a multiple of ΔT and Δt , $I_{s,s}$ and $I_{s,f}$ have to be extracted and interpolated to give the correct values. As the propagation time τ can be determined artificially, setting τ to a multiple of ΔT and Δt will prevent this. To reduce the affect brought by the unreal capacitances, τ is set to a minimum multiple of ΔT and Δt . For the sake of simplicity, the integration step of the slow subsystem ΔT is defined as an integer multiple of the fast subsystem Δt . The time delay τ is defined as $N\Delta t$.

$$\begin{cases} \tau = \Delta T \\ \Delta T = N\Delta t \end{cases} \quad (21)$$

A set of completely discrete system nodal equations can be obtained as

$$\begin{cases} G_s v_s(t) = i_s(t) - I_{h,s}(t - \Delta T) - I_{s,s}(t - \Delta T) - I_{s,f}(t - N\Delta t) \\ G_f v_f(t) = i_f(t) - I_{h,f}(t - \Delta t) - I_{f,f}(t - N\Delta t) - I_{f,s}(t - \Delta T) \end{cases} \quad (22)$$

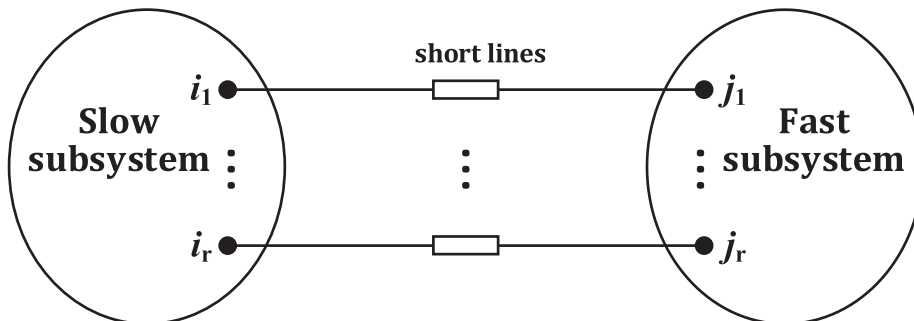


Fig. 3. Sample network for multirate simulation decomposition.

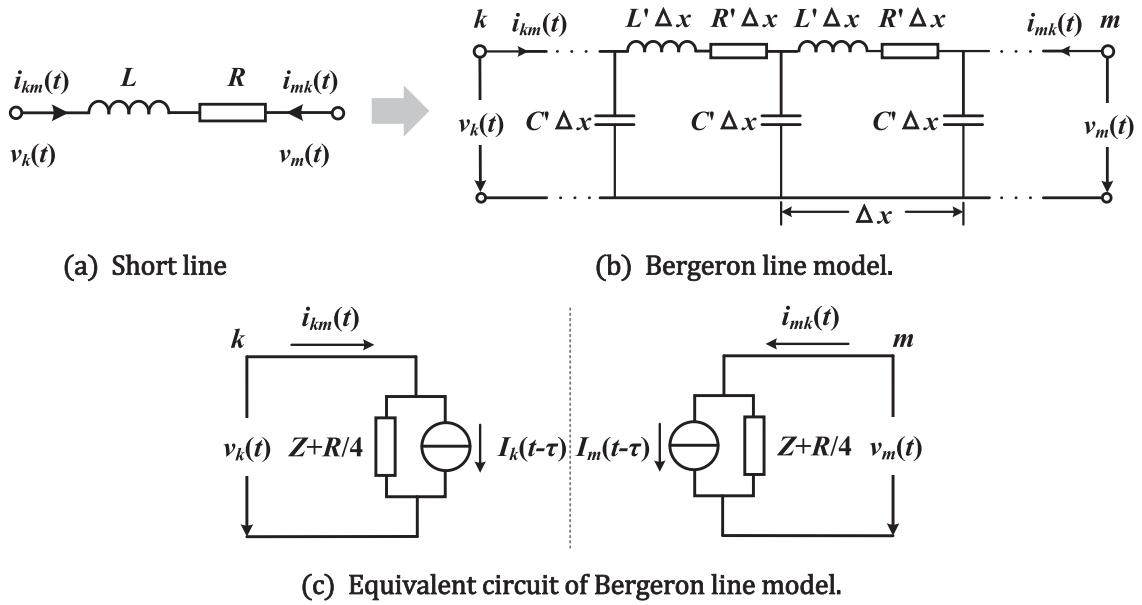


Fig. 4. Bergeron line model and its equivalent circuit model.

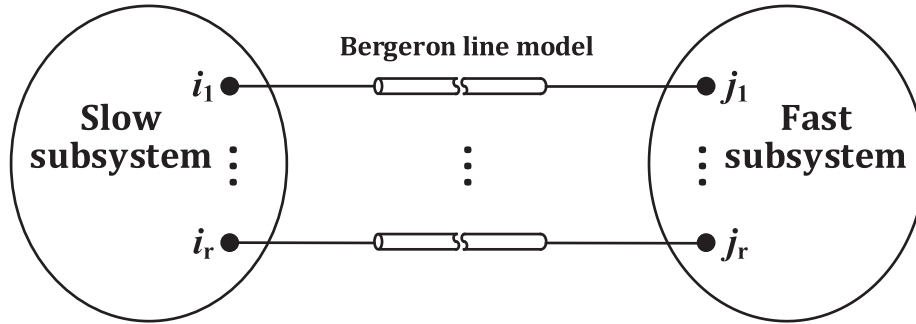


Fig. 5. Sample network decoupled by Bergeron line model.

3.2. Multirate interfacing method

At the small time steps Δt within the large step ΔT , only the fast subsystem is solved. If the slow subsystem's contribution for the solution of the fast subsystem is completely neglected, incorrect solution will result. Also, at the instant that both the two subsystems are solved, the variation of the fast subsystem solution through the entire ΔT should be considered for the solution of the slow one to reduce the probable sampling errors [17]. The contributions of both the subsystems for each other are evaluated.

(1) Multirate interface for the slow subsystem

At the small time steps Δt within the large step ΔT , the fast subsystem will be solved for N steps

$$\begin{cases} \mathbf{G}_f \mathbf{v}_f(t + k\Delta t) = \mathbf{i}_f(t + k\Delta t) - \mathbf{I}_{h,f}(t + (k-1)\Delta t) - \mathbf{I}_{f,f}(t + k\Delta t - N\Delta t) \\ - \mathbf{I}_{f,s}(t + k\Delta t - \Delta T), \quad k = 1, 2, \dots, N \end{cases} \quad (23)$$

In (23), $\mathbf{I}_{f,s}(t + \Delta t - \Delta T)$, $\mathbf{I}_{f,s}(t + 2\Delta t - \Delta T)$, ..., $\mathbf{I}_{f,s}(t + N\Delta t - \Delta T)$ are unknown variables. With the assumption that the slow subsystem is varying slowly throughout the interval ΔT , a linear interpolation of $\mathbf{I}_{f,s}(t)$ is used to calculate the unknown variables. The interpolation of $\mathbf{I}_{f,s}$ is expressed as

$$\hat{\mathbf{I}}_{f,s}((t + k\Delta t - \Delta T)) = \mathbf{I}_{f,s}(t - \Delta T) + \frac{k}{N}(\mathbf{I}_{f,s}(t) - \mathbf{I}_{f,s}(t - \Delta T)), \quad k = 1, 2, \dots, N \quad (24)$$

After substituting $\mathbf{I}_{f,s}$ by $\hat{\mathbf{I}}_{f,s}$, it is possible to evaluate the history current source of the slow subsystem without having to run through the small integration steps. Polynomial extrapolation can also be used to obtain the unknown variables [32].

Fig. 6 shows how the interfacing for slow subsystem works, for example, $\Delta T = 4\Delta t$. In Fig. 6, “●” indicates the instants when both the slow and the fast subsystems are solved. “○” indicates the instants when only the fast subsystem is solved. “○” is the interpolation point.

The detailed interfacing procedure of the slow subsystem is as

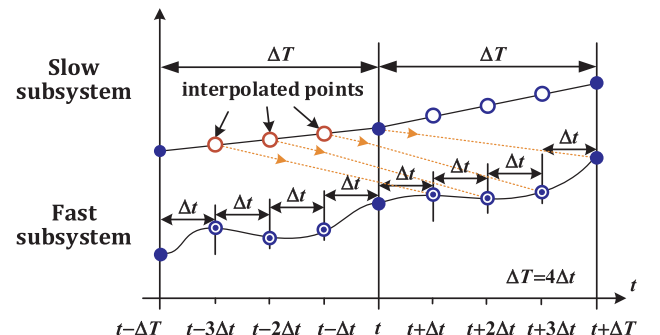


Fig. 6. Multirate interfacing for the slow subsystem.

follows:

- (i) Determine the history current sources $I_{f,s}(t)$ of the slow subsystem by (19) at t ;
- (ii) Calculate the interpolation values $\hat{I}_{f,s}(t-3\Delta t)$, $\hat{I}_{f,s}(t-2\Delta t)$ and $\hat{I}_{f,s}(t-\Delta t)$ by (24);
- (iii) Using the interpolation values together with $I_{f,s}(t)$, the fast subsystem is solved for solutions at instant $t + \Delta t$, $t + 2\Delta t$, $t + 3\Delta t$ and $t + \Delta T$ shown in Fig. 6.

Hardware design of the multirate interface for the slow subsystem is shown in Fig. 7. It realises the pipelined computation for the interpolated values. In the design, four random-access-memory (RAM) units, one read-only-memory (ROM) units and one first-in-first-out (FIFO) block are placed. RAM_h and RAM_t are used to store the values v_h and v_t to be interpolated. $RAM_{\Delta v, xN}$ and $RAM_{v, xN}$ are used to store the difference value Δv and v_t , that will be later read out for N times. N is the number of times of interpolation operations. The storage format of $ROM_{integer}$ is $\{0, 0, \dots, 0, 1, 1, \dots, 1, \dots, N-1, N-1, \dots, N-1\}$. In the format, each element is repeated g times which is the number of the pipelined values.

(2) Multirate interface for the fast subsystem

At the instant t when both the slow and fast subsystem are solved, the history current source of the fast subsystem is updated with the solution at $t - \Delta t$. It can't represent its variation through a complete ΔT from the viewpoint of the slow subsystem. An average of the history current sources of the fast subsystem through ΔT is more accurate [17].

Assuming $\Delta T = 4\Delta t$, the nodal equation of the slow subsystem can be expressed as

$$G_s v_s(t) = i_s(t) - I_{h,s}(t - \Delta T) - I_{s,s}(t - \Delta T) - I_{s,f}(t - 4\Delta t) \quad (25)$$

At the instant when the solutions are resynchronized, the history current sources are both updated as Fig. 8. For the slow subsystem, $I_{s,s}(t - \Delta T)$ accommodate the variation from $t - 2\Delta T$ to $t - \Delta T$. For the fast subsystem, it has been solved 4 times and $I_{s,f}(t - 4\Delta t)$ accommodates the variation from instant $t - 5\Delta t$ to $t - 4\Delta t$. From the viewpoint of the slow subsystem, fast subsystem's variation during Δt cannot represent the variation during ΔT . In this paper, an average of values $\hat{I}_{s,f}$ during ΔT is used to substitute $I_{s,f}$. $\hat{I}_{s,f}$ is calculated as

$$\hat{I}_{s,f}(t - N\Delta t) = \frac{1}{N} \sum_{k=1}^N I_{s,f}(t - (k-1)\Delta t - N\Delta t) \quad (26)$$

The detailed interfacing procedure of the fast subsystem is as follows:

- (i) Determine the history current sources $I_{s,f}(t - 7\Delta t)$, $I_{s,f}(t - 6\Delta t)$,

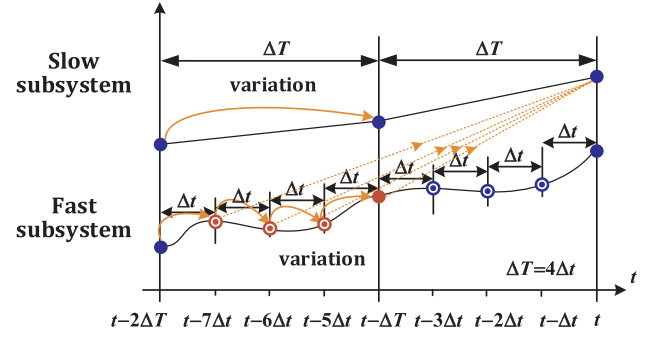


Fig. 8. Multirate interfacing for the fast subsystem.

$I_{s,f}(t - 5\Delta t)$ and $I_{s,f}(t - 4\Delta t)$ of the fast subsystem by (19);

- (ii) Calculate the average values $\hat{I}_{s,f}(t - 4\Delta t)$ by (26);
- (iii) Using the average value, the slow subsystem is solved for solutions at t shown in Fig. 8.

In Fig. 9(a), the hardware design of the interface for the fast subsystem is realised with a pipelined computation scheme. In the design, input x , in pipelined format, represents the history current sources of the fast subsystem throughout ΔT . The interface design consists of one floating-to-fixed point converter, one fixed-to-floating point converter, one accumulation unit, two FIFOs, one ROM and one RAM. $FIFO_x$ is used to rearrange the input and insert zeros in the pipeline. The core of accumulation unit is a fixed point adder in which the addition operation is forced to finish in one clock cycle. $FIFO_{sum}$ is used to select the required summation from the pipeline. The timing diagram of this operation is shown in Fig. 9(b).

3.3. Hardware design of the multirate simulator

To ensure that each FPGA can be configured to simulate any subsystem whereas the time-scale, a universal hardware design is realised in Fig. 10. It provides the full simulation functions for both network components and distributed generation units. The multiplexer in Fig. 10 is to decide whether the simulated subsystem is 'fast' or 'slow'. The subscript 'i' indicates the interface variables of the inside subsystem and 'o' indicates the variables from outside.

The hardware design comprises five modules, including electrical system solving module, control system solving module, multirate interface module, data exchange module and global control module.

- (1) In the electrical system solving module, different kinds of component models are established, such as lines, transformers, power electronic circuits, breakers and so on. The history term vector I_{hist}

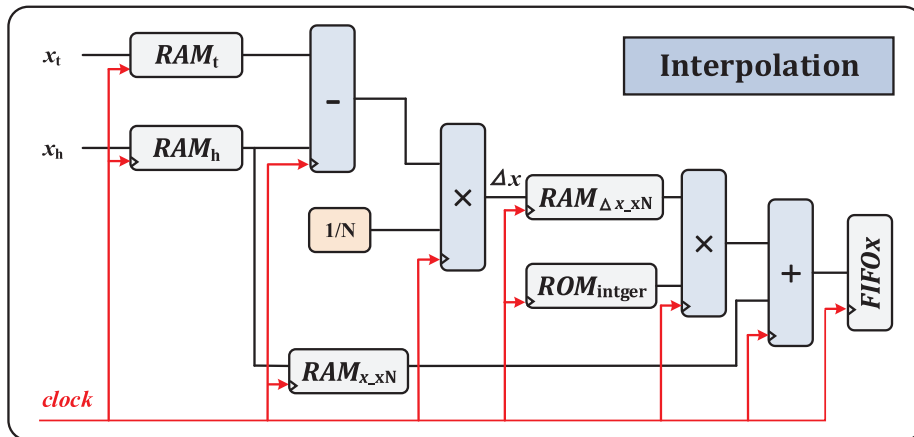


Fig. 7. Hardware design of the multirate interface for the slow subsystem.

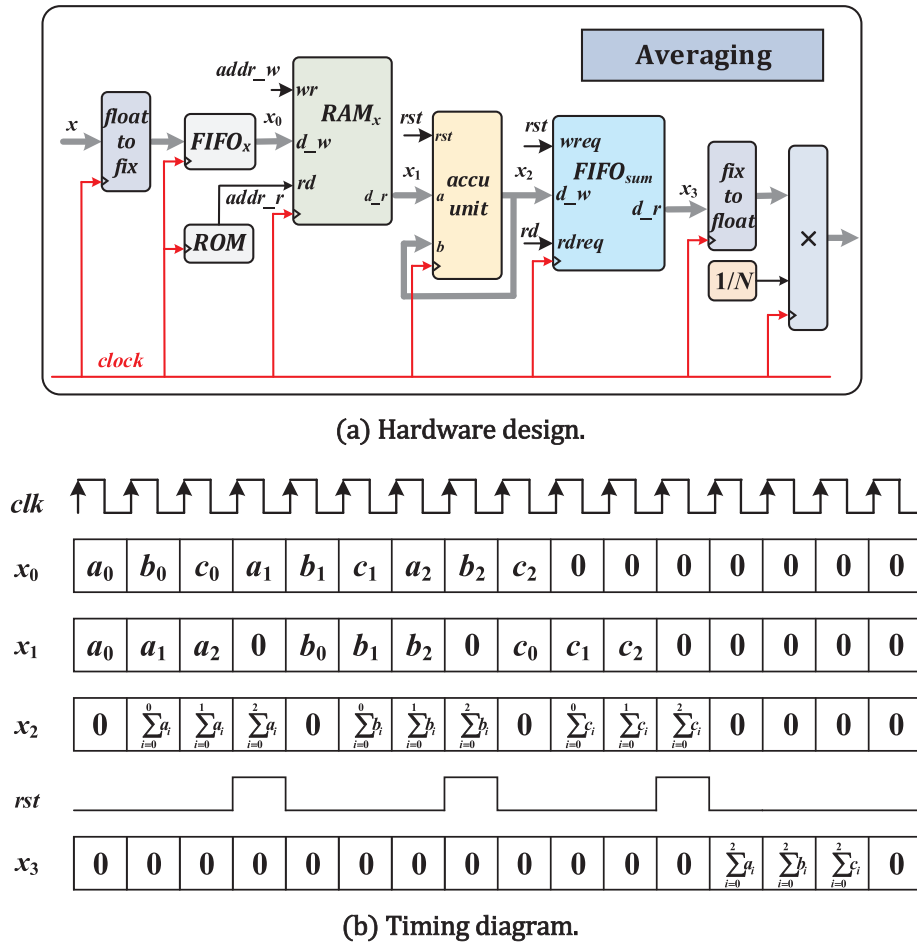


Fig. 9. Hardware design and the timing diagram of the fast subsystem interface.

is the summation of the Bergeron line model history term I_{IF} and the other network history term I_h . After the right inverse admittance matrix G^{-1} is selected by the breaker state, the network solver is executed to obtain system nodal voltages.

- (2) Control system solving module covers the basic functioning blocks, such as transfer functions, trigonometric functions, comparators, limiters, exponential functions and so on. Generally, the resulting nonlinear equations should be solved by the iterative method to acquire the correct solution. For real-time simulation, it is not suitable since the numbers of iterations are uncertain [17]. This paper adopts the pseudo-nonlinearisation method to solve the problem. The nonlinear equations, resulting from the relationships between the blocks, are transformed into linear equations by adding a time-step delay [28]. Considering that the control system is a relatively weak stiff system, the transition is appropriate.
- (3) In the multirate interface module, the inside history current sources of the interface $I_{IF,i}$ is calculated by (14). The outside history current sources $I_{IF,0}$ is calculated using the interface node voltage $v_{IF,0}$ and branch current $i_{IF,0}$. Both the interpolation and averaging operations are conducted for $I_{IF,0}$. A multiplexer is used to determine the required transformation of $\hat{I}_{IF,0}$. Then the summation of $I_{IF,i}$ and $\hat{I}_{IF,0}$ are sent to the electrical system solving module for system nodal voltages calculation. After obtaining the interface nodal voltage $v_{IF,i}$, it is sent back to the multirate interface module for next simulation step.
- (4) Data exchange module accomplishes the bidirectional data transmission between FPGAs by configuring their high-speed transceivers. The transmission data include the branch voltages and branch currents of the interface during each time-step.

- (5) Global control module sends resetting, initializing and starting signals to each module to enable it performing the designed function. It also receives the signals sent back from each module to judge if the required function has been executed properly. Especially, this module sets up a multirate synchronisation mechanism [33] to ensure that data transmission between FPGAs is in an orderly manner.

4. Case studies and analysis

In this section, real-time simulation of two modified IEEE 33-node systems with photovoltaics and energy storage are implemented on a 4 FPGA-based real-time simulator. The performance of the proposed multirate simulation method and multi-FPGA hardware design is investigated by the minimum time-step with respect to the simulated system size. Simulation results are also compared with the commercial off-line transient simulation tool PSCAD/EMTDC to validate the multi-FPGA based simulator.

4.1. Multi-FPGA-based real-time simulator

Fig. 11 shows the multi-FPGA-based real-time simulator. Four Stratix® V Edition DSP development boards from Altera® are used. Each FPGA development board provides four 2.5Gbps full-duplex optical fibre communication channels that enable the flexible connection between FPGAs. FPGA1 is selected as the master FPGA and sends synchronisation signals to the other three slave FPGAs. The slave FPGA starts simulation only if it receives the synchronisation signal and the start time of the master FPGA is pushed back by the transmission delay

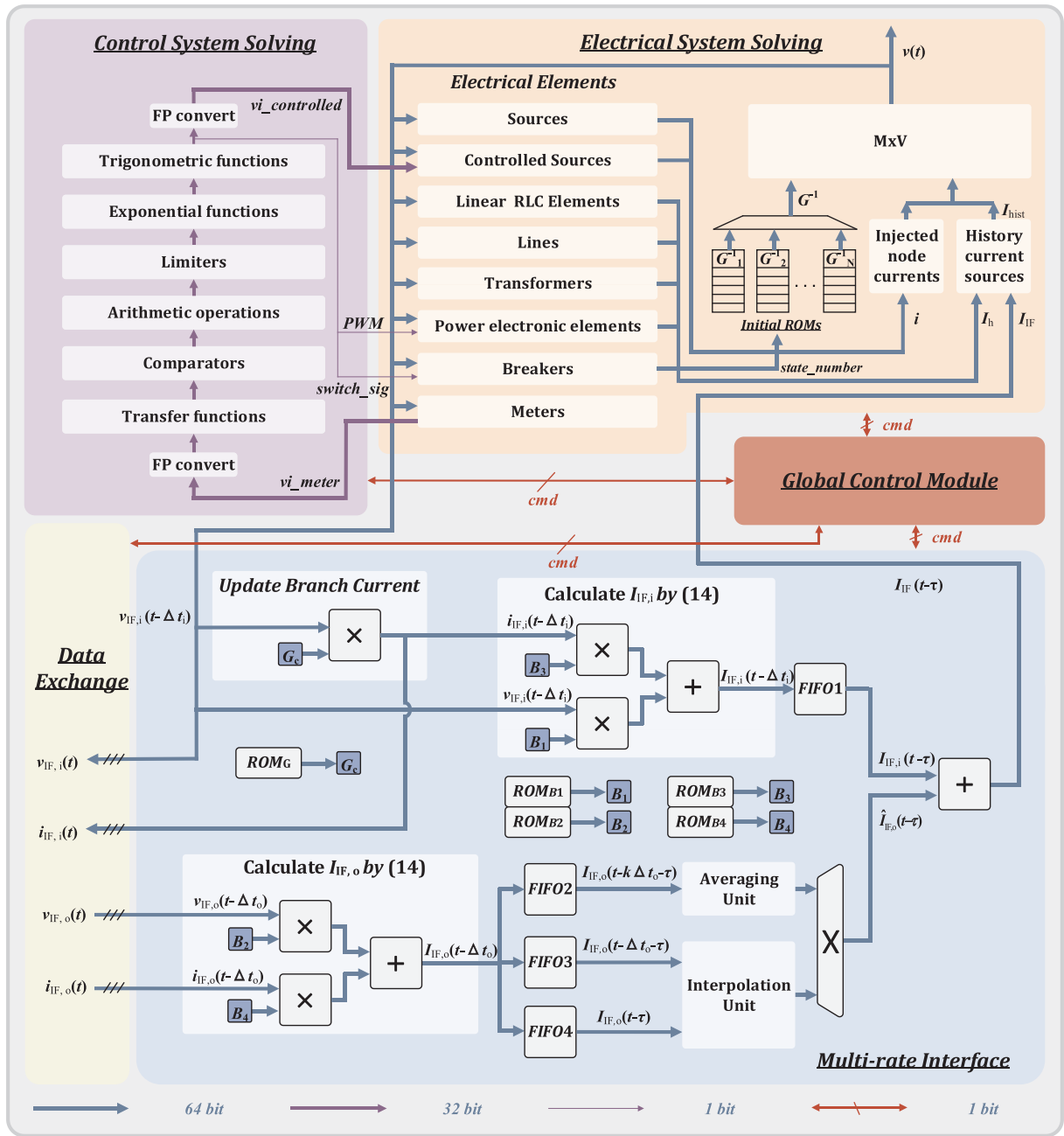


Fig. 10. Hardware design of the FPGA-based extendable real-time simulator.

of the signal. The maximum driven clock frequency for FPGAs is 125 MHz, which severely determines the execution time of the hardware design. Real-time results can be captured either by the oscilloscope via DA converter or the host computer through JTAG bluster.

4.2. Test case

To verify the performance of the FPGA-based multirate real-time simulator, two test cases are studied in this subsection.

(1) Test case 1

Test case 1 is a modified IEEE 33-node system with single PV units and energy storage, shown in Fig. 12(a). It is set to elaborate the detailed transients of the active distribution network, especially the inherent characteristics of the PV units and energy storage. The entire

system is composed of one IEEE 33-node network, two single PV units and one integrated PV/battery unit. The detailed parameters of the IEEE 33-node network are provided in [34]. The detailed structure of the integrated PV/battery unit and the PV unit are described in Appendices B and C respectively. The equivalent circuit of PV arrays and batteries are described in [35,36].

In the PV/battery unit, the PV array is connected to the DC bus via a DC/DC boost inverter. The maximum power point tracking (MPPT) control strategy [37] is adopted to ensure the maximum active power output of PV array. The battery is connected to the DC bus via a DC/DC dual-directional converter. The common coupling DC bus and the AC network are linked with DC/AC inverter. P-Q control strategy is employed to regulate both of the active and reactive power of the DC/AC inverter to meet the power demand. The two single-stage grid-connected PV units connected to node 33 and node 22 are structurally identical. In this case, the reference voltage of the DC bus in PV/battery

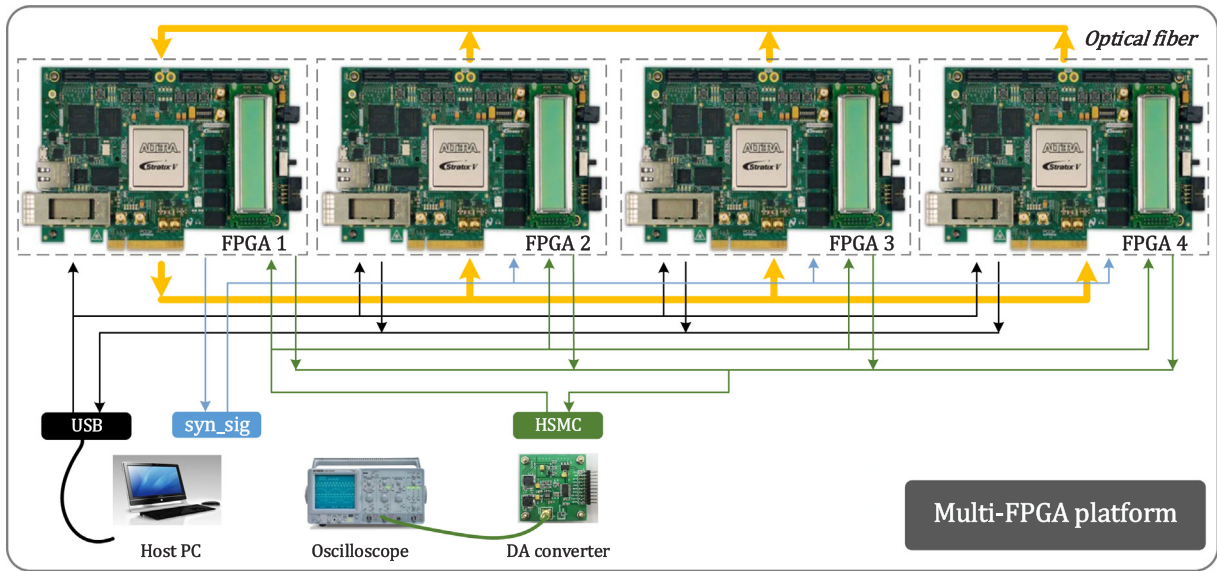


Fig. 11. The integrated simulation platform.

unit is set to 750 V, the active power demand is 6 kW, and the initial state of charge (SOC) of the battery is 60%. The entire system is decomposed into four subsystems including the IEEE 33-node network, the integrated PV/battery unit, PV unit 1 and PV unit 2. Each subsystem is simulated on a single FPGA

Table 1 displays the numbers of different types of elements for each subsystem. The entire test case includes 10 sources, 124 RLC elements, 96 lines, 21 IGBTs, 22 diodes, 28 m, 9 transformers, 9 interfaces and a breaker.

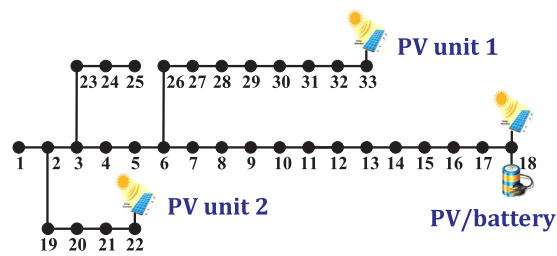
Table 2 lists the main computational resources utilized by each subsystem. As can be seen, although the simulation scale of DGs and energy storage is 13–20% of the network, the computational resources utilized is 60–75% of the network. This is due to the complicated modelling of DG and energy storage as well as the controllers for power electronic circuits, which entails more computational resources than traditional power systems. Considering that the logic resources and DSP

resources of FPGA are interconvertible, adjusting the proportion properly is a feasible way to further expand simulation scale.

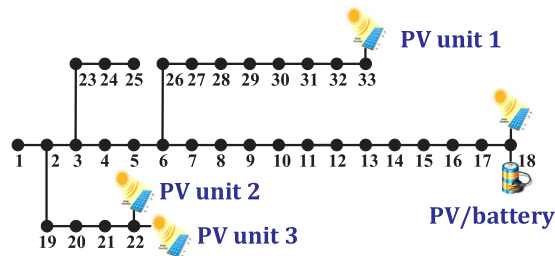
The execution time per time-step for each subsystem is shown in Table 3. It is composed of electrical system solving time t_e , control system solving time t_c , communication time t_{comm} , interpolation time t_{in} and averaging time t_{ave} . As the control system is solved in parallel with the other processes, the final execution time t_s is determined by the maximum value of t_c and $(t_e + t_{comm} + t_{in} + t_{ave})$.

According to the execution time, the synchronisation signal interval is set to 2 μ s, completing the communication, interpolation and averaging during the interval. The time-steps for PV/battery unit and the two PV units are set to 4 μ s. The time-step for IEEE 33-node network part is set to 8 μ s.

(2) Test case 2



(a) Test case 1: Modified IEEE 33-node system with single PVs and energy storage.



(b) Test case 2: Modified IEEE 33-node system with paralleled PVs and energy storage.

Fig. 12. Test cases.

Table 1
Numbers of different types of elements for each subsystem.

Subsystem	Source	RLC	IGBT	Diode	Meter	Line	Transformer	Breaker	Interface	Matrix dimension
Network	3	96	0	0	0	96	0	1	9/2	122
PV/battery	3	12	9	10	12	0	3	0	3/2	26
PV unit 1	2	8	6	6	8	0	3	0	3/2	16
PV unit 2	2	8	6	6	8	0	3	0	3/2	16

Table 2
Computational resource utilization for the 4-FPGA based hardware design.

Subsystem	Logic resources 172,600	DSP blocks 1590	Memory bits 41,246,720	Transceiver 36
Network	90%	31%	34%	3
PV/battery	68%	9%	19%	1
PV unit 1	54%	6%	18%	1
PV unit 2	54%	6%	18%	1

In ADNs, renewable sources are typically connected to a point of common coupling (PCC) with the grid by using multiple inverters for scalability and reliability reasons [38–40]. Some researchers have concerned that in such system architectures, the parallel connected inverters interact among each other and with the grid, which may bring to increased harmonic current distortion at inverters output and potentially to instability issues [41]. To explain the phenomenon resulting from the control interactions, a modified IEEE 33-node system with paralleled PV units and energy storage, shown in Fig. 12(b), is modelled and simulated on the proposed simulator. The tested system is built on the foundation of the system in Test case 1 by integrating an additional PV unit to node 22. The architecture and parameters of the added PV unit (PV unit 3) are the same with that of PV unit 1 and PV unit 2 in Test case 1. Similarly, the entire system is decomposed into four subsystems, i.e. the IEEE 33-node network, the integrated PV/battery unit, PV unit 1 and the paralleled two PV units. Each subsystem is executed on a single FPGA. The time-step for the IEEE 33-node network is 8 μ s and the others are 4 μ s. The simulation results will be elaborated in the next subsection.

4.3. Simulation result validations

Two events are simulated to understand the transients of the tested cases. The first transient event, a Phase-A ground fault occurs at the output port of the filter in PV unit 2 at 2.0 s and is then cleared after 0.2 s for both the two test cases. The second transient event is an irradiance arising from 500 W/m² to 1000 W/m² at 2.4 s. To demonstrate the multirate simulation method and the hardware design, the simulation results are captured and plotted along with PSCAD/EMTDC. The time-step for this commonly used transient simulation tool is 4.0 μ s.

(1) Test case 1

Fig. 13(a)–(d) shows the waveforms of the PV/battery unit in Test case 1. During the irradiance rise, the active power output of PV array increases from 5 kW to 10 kW while the total output active power is 6 kW. This is because the battery is to discharge the energy in irradiance

fall and store excess energy during high irradiance condition. Also, the DC voltage is kept to the desired 750 V and the peak current remains 15.0 A during the rise. The Phase-A voltage and current waveforms of PV unit 2 are plotted in Fig. 13(e) and (f). The voltage drops from 150 V peak to 0 V during the fault and raises back to 150 V peak after the fault is cleared. The Phase-A current waveform is distorted during the fault and recovers after the clearance. When the irradiance arises, the current raises from 25.0 A to 50.0 A. Fig. 14(a)–(c) shows the Phase-A voltage, current and DC voltage curves of the PV unit 1 in Test case 1 under different time-steps of 8.0 μ s, 12.0 μ s, and 16.0 μ s. Fig. 14(d) shows the relative error of the DC voltage compared with the result of PSCAD/EMTDC under 2.0 μ s as a reference.

As can be seen from Figs. 13 and 14, the results of the FPGA-based real-time simulator are very close to that of PSCAD/EMTDC, which verifies the accuracy of the proposed method and the correctness of hardware design. In contrast to PSCAD/EMTDC, the errors of the FPGA-based simulator incorporates the following aspects: (i) Part of the hardware design uses single-precision floating-point data format to decrease the consumption of logic resources and DSP blocks. (ii) The modelling approach of power electronic circuits in FPGA and PSCAD/EMTDC are different as Section 2.2 explained. (iii) In the FPGA-based simulator, different time steps are used for different network subsystems while the single time step is used in PSCAD/EMTDC. (iv) Bergeron line models are introduced to decompose the ADN, which further brings errors.

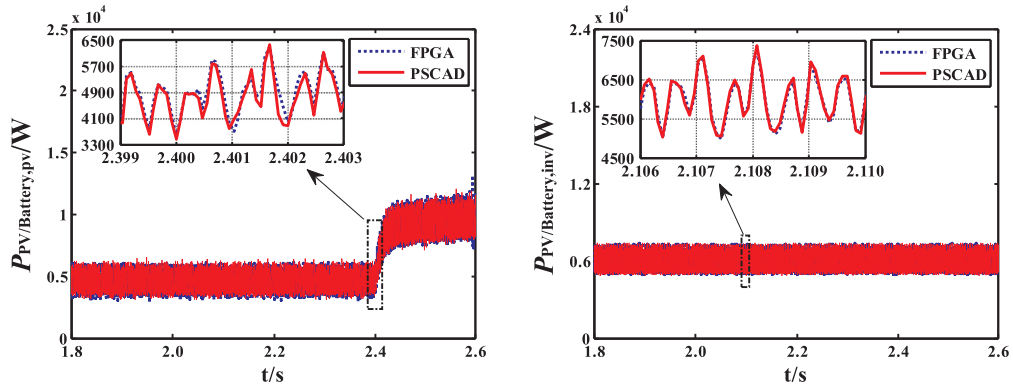
(2) Test case 2

To elaborate the phenomenon resulting from the control interaction, the waveforms of the Phase-A current and reactive power of PV unit 2 in Test case 2 are compared with that of the PV unit 2 in Test case 1, as shown in Fig. 15. It can be seen that the current is distorted since the present controller does not track the command current well that the system becomes unstable ultimately leading to a poor grid current quality. The reactive power of the PV unit 2 fluctuates with a larger range leading to a poor power factor. In terms of the industry researchers' concern, the proposed multi-FPGA-based multirate real-time simulator makes it possible for revealing the phenomena caused by the control interactions, which lays a foundation for the further study of adequate controller designing.

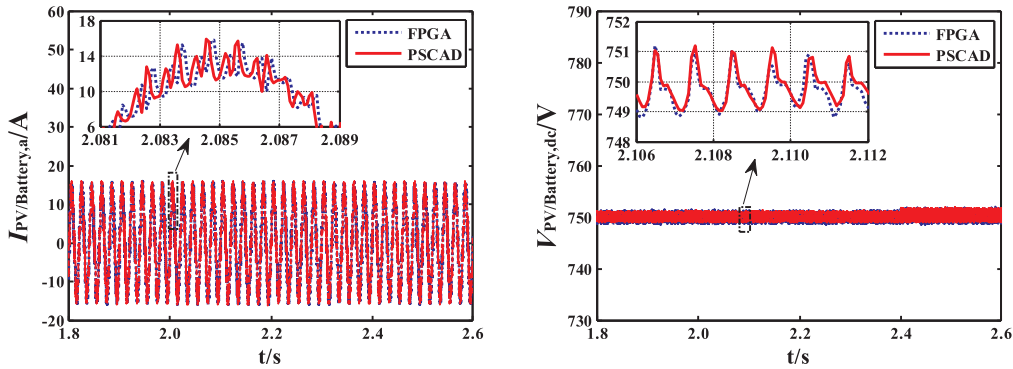
The proposed multi-FPGA-based multirate real-time simulator in this paper has been verified to reproduce the transients of ADNs with acceptable fidelity. The major functional application for the simulator is testing the physical device under real-time conditions which provides close reproduction of how the device will behave in the field. The simulator can also be used for fast developing models and rapid prototyping of novel device/concept in power and energy systems, which

Table 3
Execution time of each subsystem per time-step (μ s).

Subsystem	Electrical system, t_e	Control system, t_c	Communication, t_{comm}	Interpolation, t_{in}	Averaging, t_{ave}	Total, t_s
Network	4.640	0	0.928	0	0.528	6.096
PV/battery	1.04	2.256	0.832	0.288	0	2.256
PV unit 1	0.880	2.032	0.880	0.288	0	2.048
PV unit 2	0.880	2.032	0.896	0.288	0	2.064

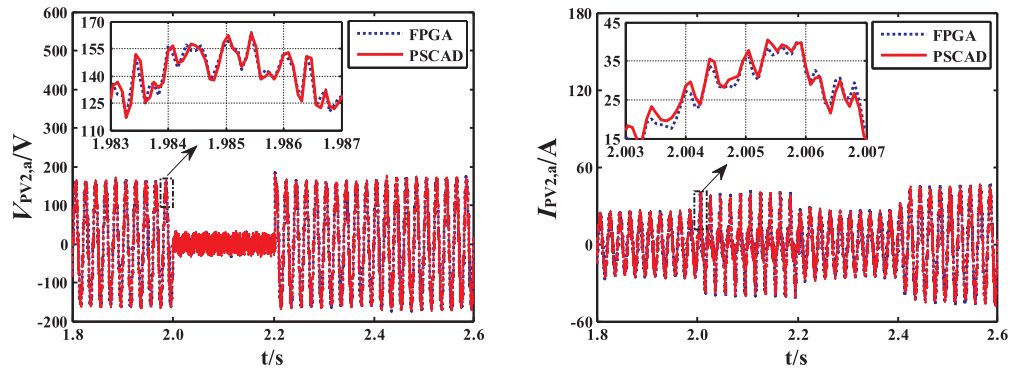


(a) Active power output of PV in PV/battery unit. (b) Active power output of PV/battery unit.



(c) Phase-A current of PV/battery unit.

(d) DC voltage of PV/battery unit.



(e) Phase-A voltage of PV unit 2.

(f) Phase-A current of PV unit 2.

Fig. 13. Simulation results of the active distribution network in Test case 1.

leads to a more efficient design process in industrial applications.

5. Conclusion

This paper presents an extendable field-programmable-gate-array-based multirate real-time simulation method for active distribution networks, which realises the high-accuracy real-time simulation for large-scale networks containing enormous high-frequency power electronic devices. The overall system is decoupled into smaller subsystems based on the time-scales. Each subsystem is mapped into an individual field programmable gate array and distinct integration steps are used. Mathematically, the subsystem is characterized by exponential difference equations which does not suffer from truncation error regardless of

the time-step sizes and thus accurate time domain solution is provided. A multirate interfacing method has been proposed to deal with the interface between the subsystems. The hardware design of the field-programmable-gate-array-based extendable real-time simulator is then achieved. Two modified IEEE 33-node systems with photovoltaics and energy storage have been simulated on the platform. Accuracy and stability of the multirate interface are demonstrated in comparison with PSCAD/EMTDC. As the hardware design of the simulator is fully extendable, employing more field programmable gate arrays will enrich the computational resources and larger system scale and faster calculation speed will be achieved.

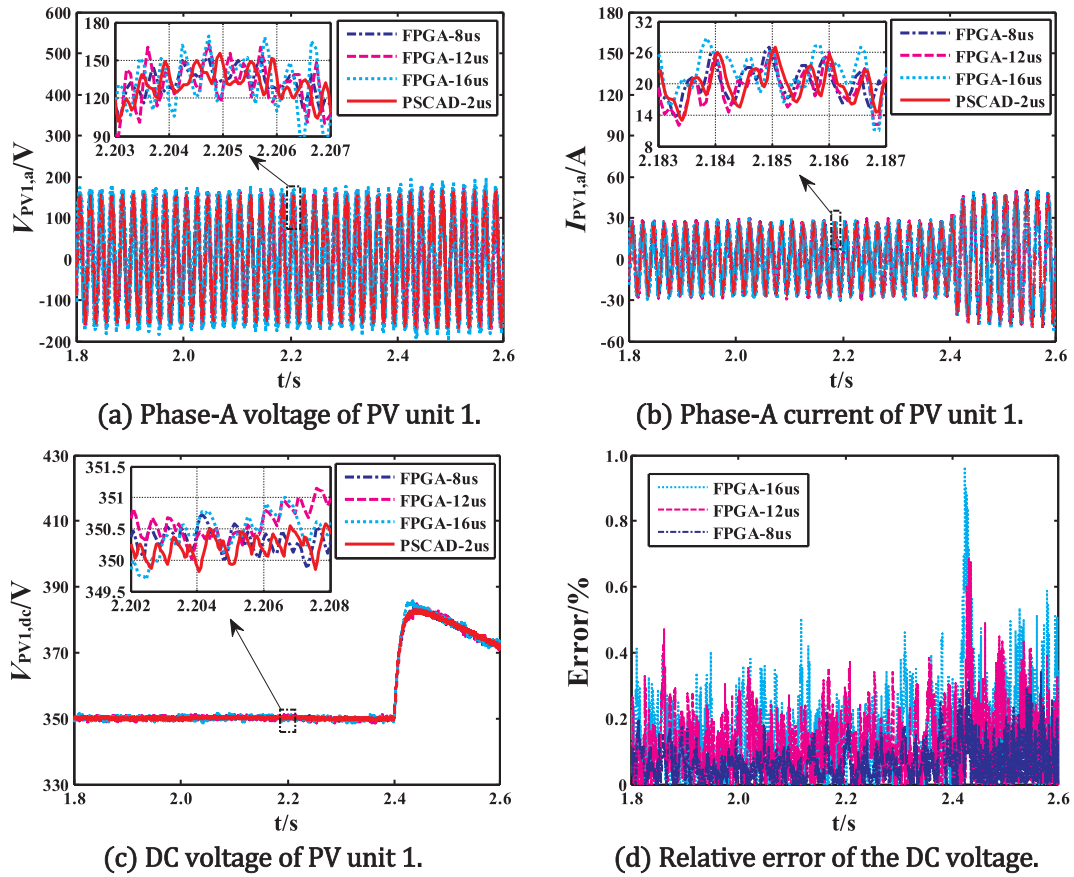


Fig. 14. Simulation results of Test case 1 and the relative error under different time-steps.

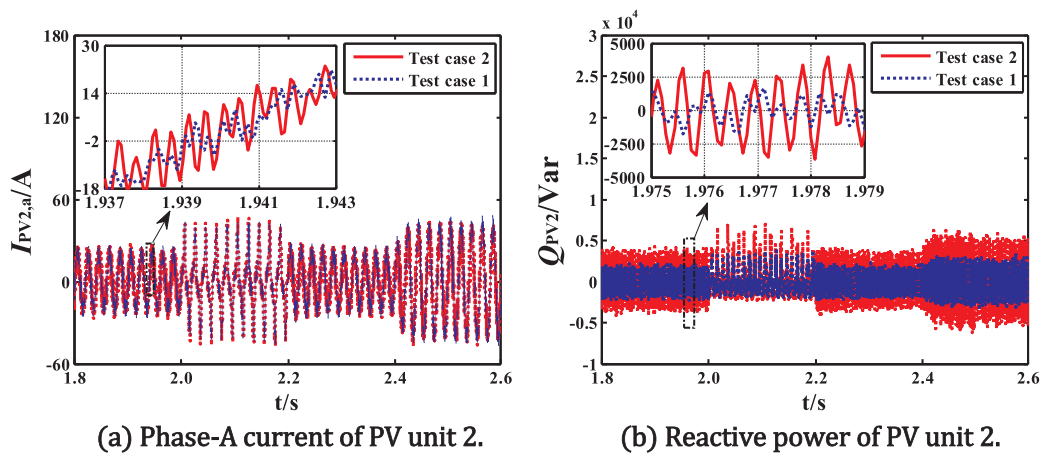


Fig. 15. Comparison of the simulation results of PV unit 2 in the two test cases.

Acknowledgements

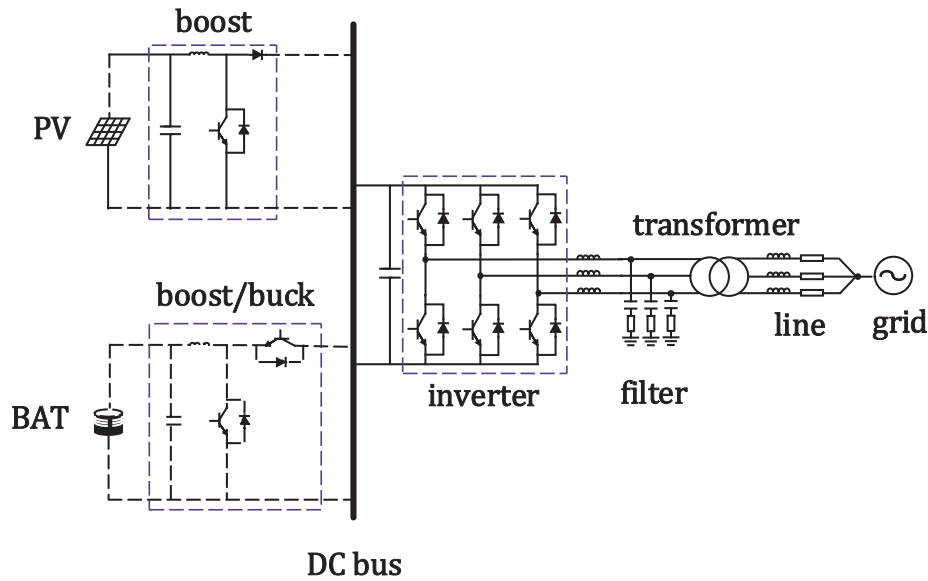
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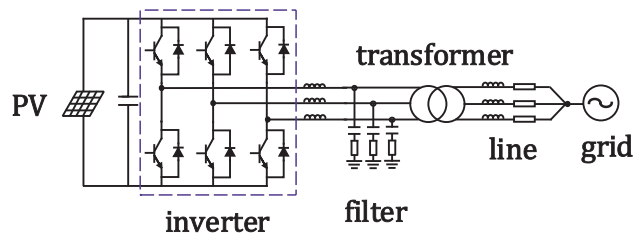
Appendix A. Exponential form terms for different branch types

Branch type	G_{eq}	A_1	A_2	A_3
R	$1/R$	0	0	0
L	$\Delta t/2L$	$\Delta t/2L$	1	0
C	$2C/\Delta t$	$-2C/\Delta t$	-1	0
RL	$\frac{(1 - e^{-\Delta t R/L})}{R}$	0	$e^{-\Delta t R/L}$	0
RC	$\frac{C(1 - e^{-\Delta t/(RC)})}{\Delta t}$	$-\frac{C(1 - e^{-\Delta t/(RC)})}{\Delta t}$	$e^{-\Delta t/(RC)}$	0
R//L	$\frac{1}{L(1 - e^{-\Delta t R/L})}$	$-\frac{e^{-\Delta t R/L}}{L(1 - e^{-\Delta t R/L})}$	1	0
LC	$\frac{C(2 - 2\cos(\Delta t\sqrt{1/(LC)}))}{\Delta t}$	$-\frac{C(2 - 2\cos(\Delta t\sqrt{1/(LC)}))}{\Delta t}$	$2\cos(\Delta t\sqrt{1/(LC)})$	-1

Appendix B. Detailed structure of the PV/battery unit



Appendix C. Detailed structure of the PV unit



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